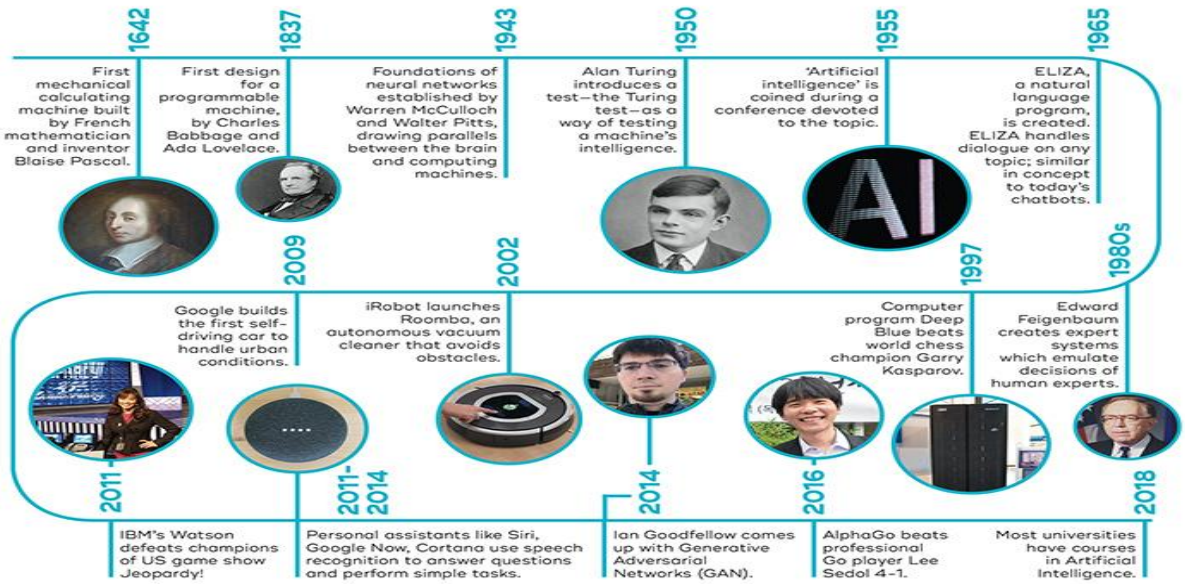
INDIVIDUAL TASK 1

Research and present a timeline showing major milestone in AI history



1. Introduction

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines capable of performing tasks that typically require human intelligence, such as learning, reasoning, problem-solving, perception, and language understanding. The development of AI did not happen overnight; instead, it evolved gradually through theoretical foundations, algorithmic breakthroughs, increased computing power, and the availability of large datasets.

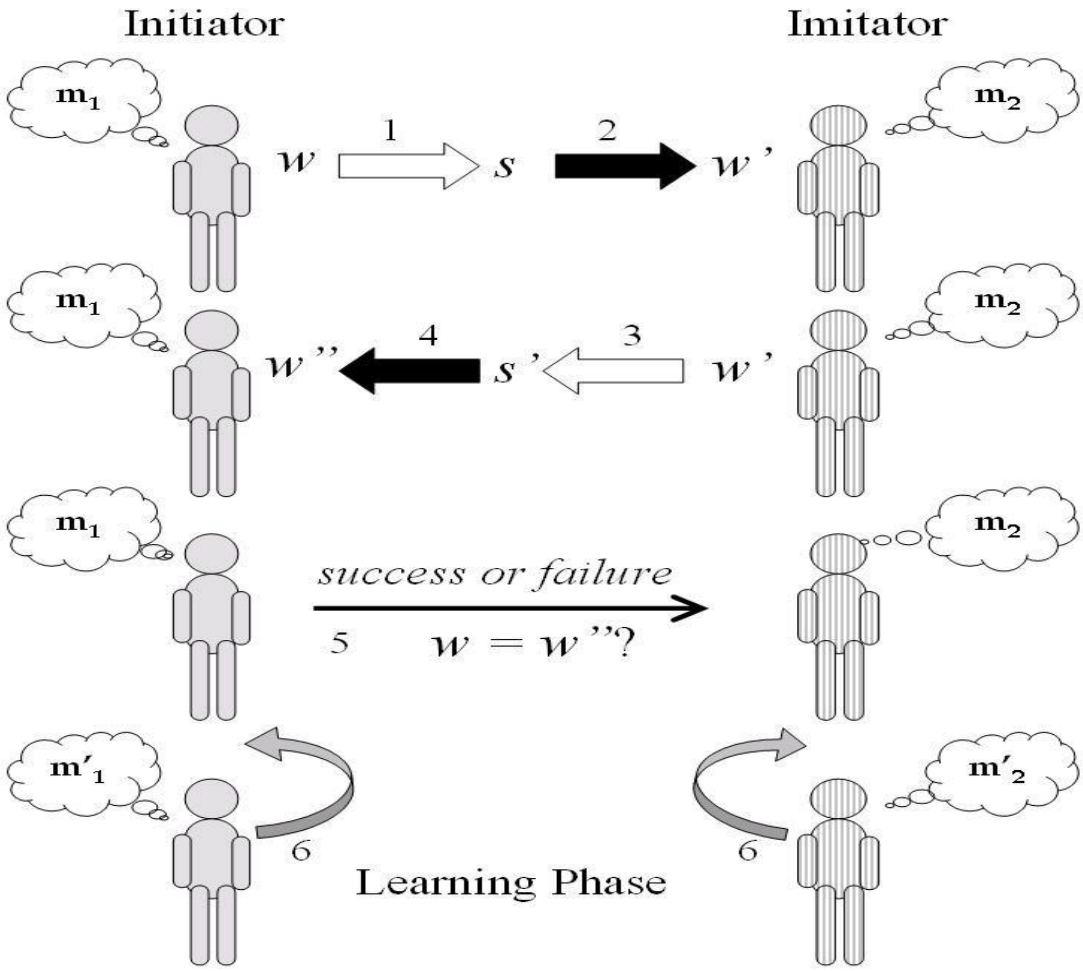
This report presents a **chronological timeline of major milestones in AI history**, explaining how each phase contributed to the growth of intelligent systems used today in areas like healthcare, finance, transportation, education, and entertainment.

Timeline of Major Milestones in AI History

- 1943 – Artificial Neurons and the Birth of AI Theory
- Walter Pitts proposed the first mathematical model of artificial neurons.
- Their model showed how neurons could be represented using logic and mathematics.

This work became the foundation of neural networks and modern deep learning.

1950 – The Turing Test



- published the paper *"Computing Machinery and Intelligence"*.
- He introduced the **Turing Test**, which evaluates a machine's ability to exhibit human-like intelligence through conversation.
- If a human cannot distinguish a machine from another human, the machine is considered intelligent.

Impact:
Provided the **first practical benchmark** for machine intelligence.

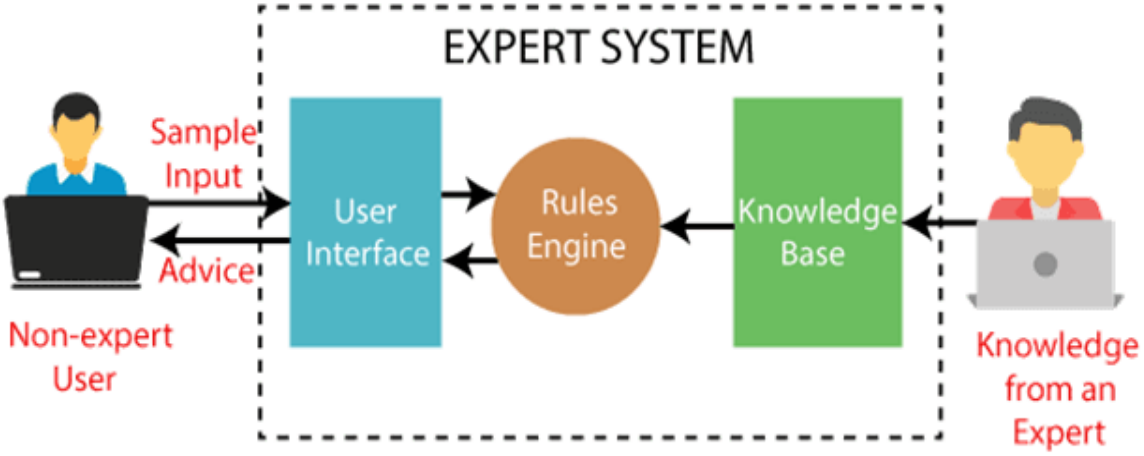
1956 – Dartmouth Conference (Birth of AI as a Field)

- Organized by John McCarthy, along with other researchers.
- The term "Artificial Intelligence" was officially coined.

- Researchers believed human-level intelligence could be achieved within a few decades.

1960s–1970s – Early AI Programs and Expert Systems

- Development of symbolic AI programs like ELIZA, an early chatbot.
- Emergence of expert systems, which used rule-based logic to mimic human experts in specific domains.
- Systems relied heavily on if-then rules created by humans.



1980s – AI Boom and Expert System Commercialization

- Expert systems were adopted in industries like medicine and manufacturing.
- AI received large investments from governments and corporations.
- However, systems were expensive and difficult to maintain.

Late 1980s–1990s – AI Winter

- Expectations from AI were not met.
- Funding and interest declined sharply.
- Many AI projects were abandoned.

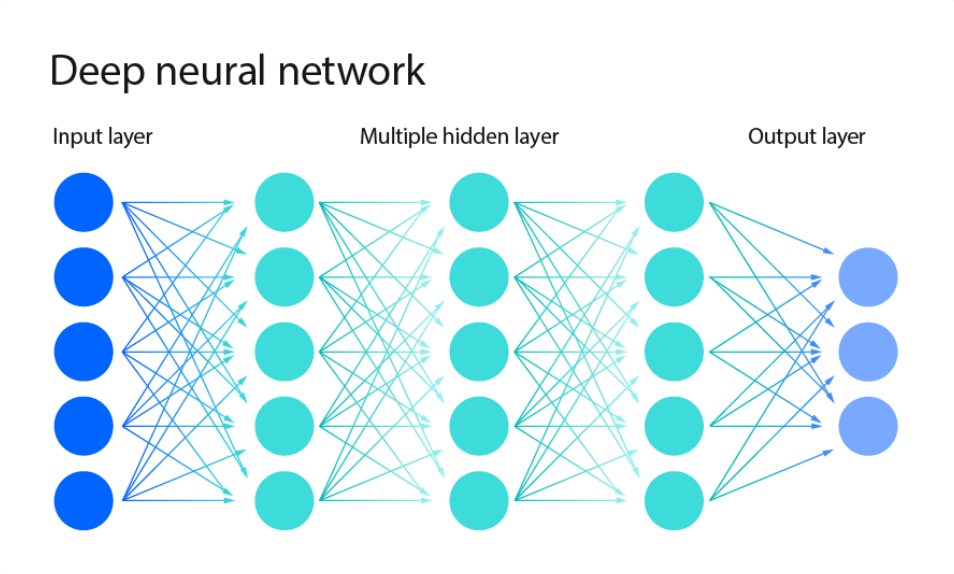
1997 – Deep Blue Defeats World Chess Champion

- IBM 's **Deep Blue** defeated **Garry Kasparov**.
- Used brute-force computation and evaluation functions.

- First time a machine defeated a world chess champion.

Impact:
Proved machines could outperform humans in complex, rule-based games.

2012 – Deep Learning Breakthrough



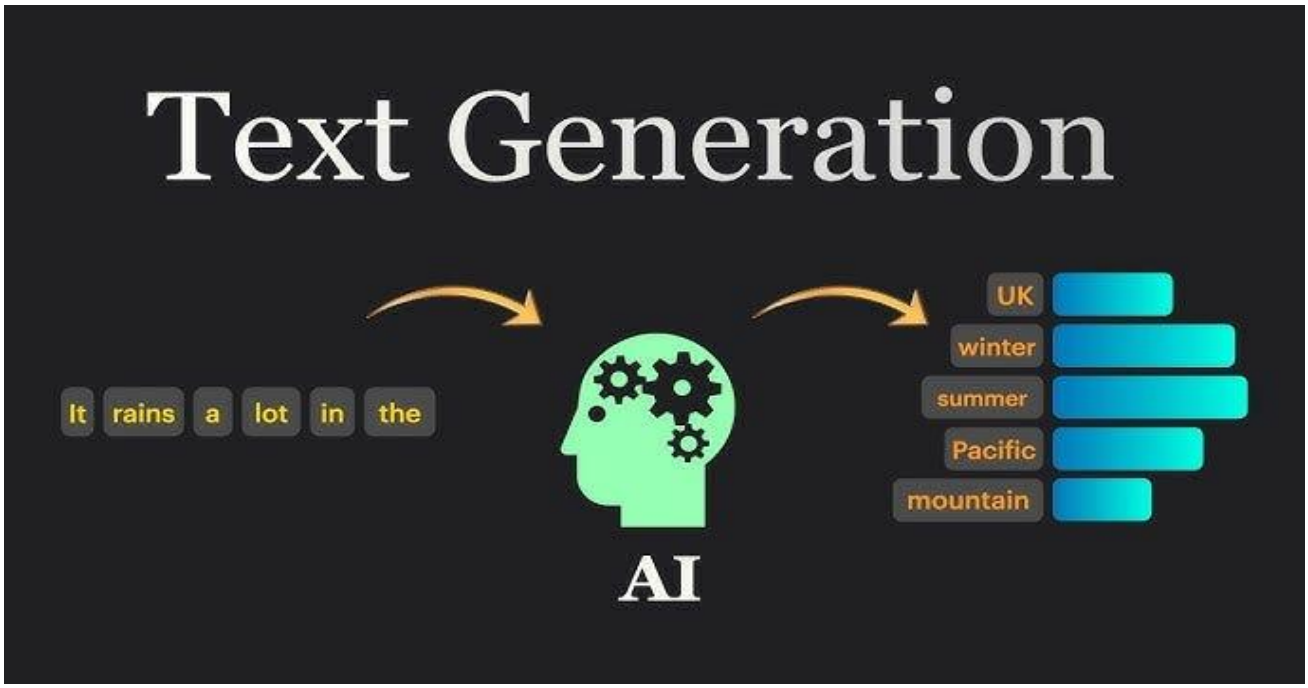
- and his team won the ImageNet competition.
- Used deep **convolutional neural networks (CNNs)**.
- Achieved dramatic improvement in image recognition accuracy.

Impact:
Sparked the **modern AI revolution** powered by deep learning.

2016 – AlphaGo Defeats World Go Champion

- Combined deep learning with reinforcement learning.
- Go was considered far more complex than chess.

Impact:
Showed AI's ability to learn strategies beyond human programming.



2020s – Generative AI and Large Language Models

- Rise of large language models (LLMs) capable of generating human-like text.
- Applications include chatbots, coding assistants, image generation, and healthcare analysis.
- AI systems now learn from massive datasets using billions of parameters.

Impact:
AI became widely accessible and integrated into everyday life.

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3. Conclusion

The history of Artificial Intelligence shows a journey from theoretical ideas to real-world intelligent systems. Early research focused on logic and rules, while modern AI relies heavily on data, machine learning, and deep neural networks. Despite periods of slow progress like the AI Winter, continuous innovation and improved computing power have driven AI forward.

Today, AI is transforming industries and society, and future advancements promise even greater impact, making AI one of the most important technologies of the modern era.